LESSON PLAN

AIRCRAFT MAINTENANCE ORDER (1)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Loading of aircraft and determination of center of gravity, effects of variation of CG on load arm
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Loading of aircraft and determination of center of gravity, effects of variation of CG on load arm
5. Specific Objectives	1.1.5.1 Trainee will be able to understand modification, life categories and flight-testing effect of CG variation and wartime maintenance procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	CG Calculation Exercise using Load Sheet; Safety: Verify zero fuel weight; Supervision: Instructor checks arithmetic
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the standard definition of Center of Gravity?
(b) Why is the CG position critical for flight safety?
(c) How does loading affect the aircraft's load arm?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Modification and life categories • Flight-testing effect of CG variation • Wartime maintenance procedure authority	L & Ds	PPT, IB	15 min
(b)	• Determination of CG using load sheets • Calculation of moments and total weight • Impact of CG shift on load arm	L & Ds	PPT, PDF	20 min
(c)	• Effects of CG variation on stability • Limits of allowable CG range • Consequences of exceeding CG limits	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between CG variation and load arm changes.
(b) Describe the flight-testing requirements for significant CG modifications.

12. Home Assignment and Follow Up	Prepare a brief note on the authority and schedule for wartime maintenance procedures as covered in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wartime maintenance, procedure, authority to employ war time maintenance procedure schedule maintenance, component life defect rectification and repair, BDR recovery to peace time documentation	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min